

Mastering the language that powers modern technology

Python in 100 pages



Kenan Bošnjak

Python in 100 pages is not an ordinary handbook. It's your gateway into the world of programming without complications or unnecessary explanations.

Instead of thick, exhausting books, here you have a short and clear guide that takes you from your very first lines of code all the way to writing your own programs. Each chapter is easy to read and immediately teaches you through practical examples you can try on your own computer.

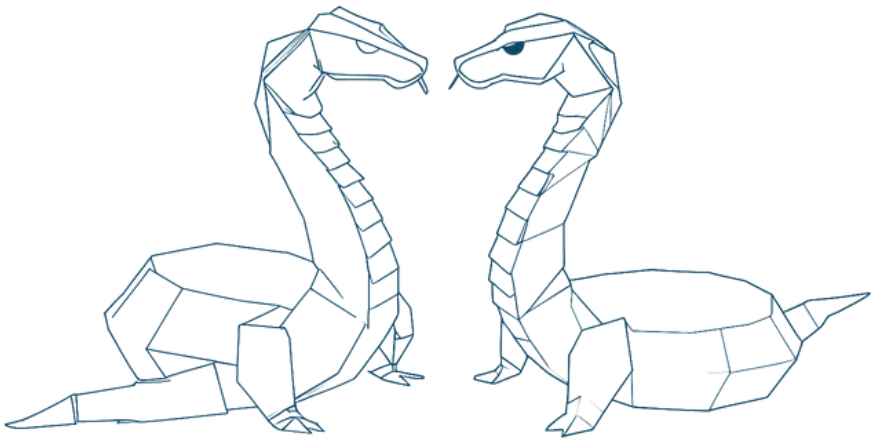
Everything is explained in simple language, without technical jargon or confusing terms. If you know how to turn on a computer, you're ready to start.

This isn't the kind of book you'll leave unread on a shelf to gather dust. This is the book that will help you write your first lines of code, understand how programs work, and open the door to a whole new world of possibilities.

If you've ever wanted to learn programming but didn't know where to start—**this is the right book for you.**



Python in 100 pages



KB

INTRODUCTION

This book is your quick introduction to the world of programming with **Python**. Without unnecessary theory, just what you need to write working programs right away. On every page, you'll find concrete examples, tasks, and tricks that both beginners and professionals can use. You'll learn through practice, mistakes, and solutions, just as **Python** is actually used. Whether you've never written a line of code or already have some knowledge, here you'll find everything you're missing: clear examples, practical skills, and the confidence to write programs yourself. By the end, you'll understand that programming isn't just about writing commands for the computer, but a method for solving problems, speeding up everyday tasks, and turning ideas into reality. Welcome to the world of programming, which will open up completely new possibilities for you.

CONTENTS:

INTRODUCTION TO PYTHON	4
Why Exactly Python?	6
BASIC SYNTAX	11
Comments	19
Variables	21
Data Types	28
Properties of Variables	31
Integer	31
Float	32
Complex	34
String	35
Boolean	38
Operators	40
Arithmetic operators	42
Assignment operators	43
Comparison operators	44
Logic operators	45
Bitwise operators	46
Membership operators	47
Identity operators	48
CONTROL FLOW	49
Conditional Statements	51
Loops	56
While Loop	57
For Loop	58
Break and Continue	59

DATA STRUCTURES	62
Lists	64
Tuples	67
Sets	69
Dictionaries	71
FUNCTIONS	74
Functions in Python	76
Lambda Functions	80
Higher-Order Functions	82
OBJECT-ORIENTED PROGRAMMING	85
Classes and Objects	86
Attributes and Methods	87
Inheritance and Polymorphism	89
ADVANCED TOPICS	92
Working with Files	93
Errors and Exceptions	99
FINAL PROJECT	101
Step-by-Step Guide	103

INTRODUCTION TO PYTHON

Imagine a language that lets you create almost anything you want — from a simple program that makes your everyday life easier to versatile software used by millions of people worldwide. That language is **Python**.

Python is a multipurpose programming language developed in 1991 by **Guido van Rossum**. Its main philosophy is based on simplicity, readability, and efficiency, making it one of the **most popular programming languages** today.

Today, it is used equally by beginners taking their first steps in the world of programming and experienced professionals developing complex and sophisticated systems. Python is used in almost all areas, from web development, data analysis, and artificial intelligence to automatic data processing, game development, and even quantum simulations.

Its strength lies in combining accessibility for new programmers with the **power** and **flexibility** that professionals need. Whether you want to automate everyday tasks, develop innovative applications, or analyze millions of data sets, Python provides you with the tools and libraries to make it happen.

Why Exactly Python?

Most programming languages have a reputation for being **difficult at first**. They make you feel like you're learning a foreign language with complicated rules and strange symbols.

Python is different. Its rules are clear and logical, so even without programming experience, you'll see results after just a few lines of code. We'll write a simple program in the C language. (*The C language is often used as an introduction to more complex programming languages. Its syntax is somewhat more complex, but it allows for faster and more direct work with the computer architecture.*)

Then we will implement the same program in Python to illustrate how much simpler and clearer the Python syntax is compared to most commercially available **programming languages** today.

Example: *Write a program that reads the names of users and outputs them in uppercase letters:*

Language C:

```
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
int main() {
    int n;
    (void)printf("How many names? "), (void)scanf("%d", &n);
    char **names = malloc(n * sizeof(char *)); if(!names) return 0;
    for (int i = 0; i < n; i++) {
        char buffer[100]; (void)scanf("%99s", buffer);
        names[i] = malloc(strlen(buffer)+1); if(!names[i]) return 0;
        strcpy(names[i], buffer);
    }
    for (int i = 0; i < n; i++) {
        for (char *p = names[i]; *p; p++) putchar(toupper(*p));
        putchar('\n');
        free(names[i]);
    }
    free(names);
    return 0;
}
```

Python:

```
n = int(input("How many names? "))
names = []
for i in range(n):
    name = input("Enter a name: ")
    names.append(name)
for name in names:
    print(name.upper())
```

As you can see, Python achieved the same result with **significantly fewer lines of code** than in the C language. In this example, the number of lines has been halved, while the code remains clear and intuitive.

Don't be discouraged if the syntax or functionality of Python seems confusing at first. That's completely normal. Everyone who writes **complex applications** with thousands of lines of code today started out at the beginning, looked at simple examples, and wondered, "How will I ever understand this?"

The **advantage of Python** is that once you have mastered the basic elements of syntax and programming logic, you can create functional programs very quickly and easily. In the beginning, these will be short and simple tasks, but over time you will create applications that contain hundreds or even thousands of lines of code without losing track of or understanding what you have written. More importantly, you will learn to write your code in a clear, organized, and easily maintainable way, which is a key skill for **any good programmer**.

Programming is not just “writing commands for the computer.” It is the ability to solve problems, think one step ahead, and develop solutions that can grow and evolve along with your ideas. What may seem like **magic** to you now will soon become part of your daily routine. When you write your first program that completes a task in seconds that would have taken you hours or days to do manually, you will understand the true power of programming, and **Python** is the perfect companion on this journey.

Interesting facts about Python:

The name does not come from a snake! Python was named after the British comedy group Monty Python. The first version was released in 1991 and is still maintained and developed today. NASA uses Python to manage certain systems and for data analysis. Google, YouTube, Instagram, Spotify—they all use Python in the background. It is one of the few languages that is equally popular with beginners and professionals alike.